Rick Mathews, PhD

I am an MD/PhD candidate at the Oregon Health & Science University. I have the privilege of being funded throughout my training by competitive NIH grants. My current research is establishing a missing and critical developmental baseline for coarctation of the aorta and is funded by the philanthropic arm of the Doernbecher Children's Hospital at OHSU. I have also contributed first-author studies across cardiovascular and thrombotic pathology, and my healthcare experience is further enriched by electronic health record project management at Epic Systems and healthcare focused venture capital operations.

EDUCATION

Doctor of Medicine (MD) 2018 – 2027 (expected)
Oregon Health & Science University Portland, OR

- Fully funded by the competitive NIH Medical Scientist Training Program and an F30 grant from the NHLBI.
- I established the cardiothoracic and vascular research repositories, directing consented and IRB approved surgical resections toward translational studies to resolve unaddressed pathologic knowledge gaps in congenital heart disease.
- Elected to the Alpha Omega Alpha Honor Medical Society (2026).

Doctor of Philosophy (PhD), Biomedical Engineering 2020 – 2024
Oregon Health & Science University Portland, OR

- Secured over $300,000 in funding from a nationally competitive NIH F30 MD/PhD fellowship, institutional Pulmonary and Critical Care grant, and the Friends of Doernbecher to investigate cardiovascular and thrombotic pathology, culminating in first-author publications that establish a missing developmental baseline for coarctation of the aorta and delineate novel mechanisms of deep vein thrombosis.

Bachelor of Engineering (BE), Biomedical Engineering 2013 – 2017
Macaulay Honors College at The City College of New York New York, NY

- Demonstrated an early and continuous pattern of securing research funding through the Wallace H. Coulter Award for Outstanding Undergraduate Research, complementing a full-tuition Macaulay Honors scholarship.

PEER-REVIEWED PUBLICATIONS: Full Profiles on ORCID, Google Scholar, and PubMed

1. **Mathews R**, et al. Territory mapping the ductal-aortic junction throughout gestation is a critical and missing baseline for coarctation of the aorta. *Am J Physiol Heart Circ Physiol*. 2026.
2. **Mathews R**, et al. Role of platelet count in a murine stasis model of deep vein thrombosis. *Platelets*. 2024.
3. **Mathews R**, et al. AMP-dependent protein kinase regulates endothelial heparan sulfate expression in response to an inflammatory stimulus under arterial shear stress. *Biochem Biophys Res Commun*. 2024.
4. **Mathews R**, et al. Venous thromboembolism: diagnostic advances and unaddressed challenges in management. *Curr Opin Hematol*. 2024.

EXPERIENCE

Platform & Portfolio Company Support 2021 – 2022
Groove Capital Minneapolis, MN

- Led the development of a comprehensive key progress indicator (revenue growth, customer acquisition cost, churn) tracking system bundled with Carta's fund administration software for 28 portfolio companies. This initiative replaced native tracking to significantly enhance performance analysis and limited partner investor transparency.

Director & Member, Board of Trustees 2018 – 2021
The Marquam Hill Cooperative Portland, OR

- Directed operations for a 23-unit affordable student housing cooperative at OHSU with over $200,000 in its annual budget, consistently sustaining above 90% occupancy.

Project Manager, Pathology Applications 2018
Epic Systems Verona, WI

- Managed implementations of clinical and anatomic pathology software (Beaker) across major healthcare networks, including Alberta Health Services in Canada and Cincinnati Children's Hospital.